



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

APRIL 2010

TASK #2

YEAR 12

Mathematics Ext 1

General Instructions:

- Reading time—5 minutes.
- Working time—90 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may NOT be awarded for messy or badly arranged work.
- Answer in simplest exact form unless otherwise stated.

Total marks—72 Marks

- Attempt questions 1–6.
- The mark-value of each question is boxed in the right margin.
- Start each NEW section in a separate answer booklet.
- Hand in your answer booklets in 3 sections:
Section A (Questions 1 and 2),
Section B (Questions 3 and 4),
Section C (Questions 5 and 6),

Examiner: Mr D. Hespe

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Section A

Marks

Question 1 (12 points)

(a) (i) How many four-letter arrangements can be made from the letters IOLS? 1

(ii) In the Herald's *Target* competition, arrangements ending in *S* are not allowed. How many four-letter *Target* arrangements can be made from the letters IOLS? 1

(b) Prove that a line from the centre of a circle to the midpoint of a chord is perpendicular to the chord. 3

(c) Use the graph of $y = \sin x$ to illustrate why 2

$$\int_{-1}^1 \sin x \, dx = 0.$$

(d) Evaluate $\lim_{x \rightarrow 0} \left\{ \frac{\tan 4x}{7x} \right\}$. 1

(e) (i) Differentiate $x \log_e x$. 1

(ii) Hence integrate $\log_e x$. 2

(f) On a certain railway line, there are eleven railway stations at which a train can stop. The rail authority needs to print tickets for travel between every possible pair of stations on the line. How many different one-way tickets must be printed if the ticket specifies which direction the passenger is travelling? 1

Question 2 (12 points)

(a) Find

(i) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \tan^2 x \, dx,$

2

(ii) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin^2 x \, dx.$

2

(b) (i) Prove by mathematical induction that

4

$$\sum_{r=1}^n \frac{1}{r(r+1)} = \frac{n}{n+1}.$$

(ii) What can you say about

1

$$\lim_{n \rightarrow \infty} \left(\sum_{r=1}^n \frac{1}{r(r+1)} \right)?$$

(c) Using one iteration of Newton's Method and a first approximation of $x_0 = 0.7$, find, correct to three decimal places, a second approximation to

3

$$y = \sin^3 x - 0.25.$$

Section B

(Use a separate writing booklet.)

Marks

Question 3 (12 points)

- (a) By writing $\cot x$ as $\frac{\cos x}{\sin x}$, evaluate

2

$$\int_{\pi/6}^{\pi/3} \cot x \, dx.$$

- (b) Given that a team of five players is to be selected from a group of ten boys, find the number of teams that contain

(i) at least one of the two best players,

2

(ii) no more than one of the three youngest players.

3

- (c) (i) Show that $e^x = 3x + 2$ has a solution between 2 and 2.5.

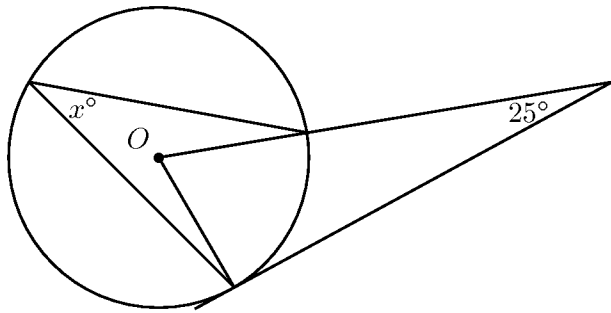
2

(ii) Hence use “halving the interval” to find, correct to one decimal place, a solution in the interval $[2, 2.5]$.

3

Question 4 (12 points)

(a)



Find the value of x , giving reasons.

3

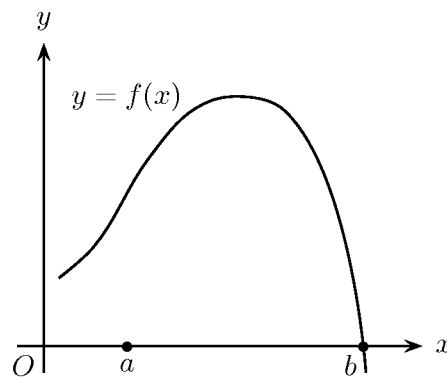
(b) Prove that $7^{n+1} + 3^n$ is divisible by 4 for all positive integers n .

4

(c) A team of three people is to be chosen from six men and five women by putting the eleven names in a hat and drawing out three simultaneously at random. Find the probability that the team will be of mixed sex.

3

(d)



Consider the above graph of $y = f(x)$. The value a shown on the axis is taken as the first approximation to the solution b of $f(x) = 0$. Is the second approximation obtained by Newton's method a better approximation to b than a is? Give a reason for your answer.

2

Section C

(Use a separate writing booklet.)

Marks

Question 5 (12 points)

(a) Differentiate with respect to x

(i) $\log_e(\cos x)$,

2

(ii) $(x + 1)e^{-x}$.

2

(b) (i) Differentiate $x + \log_e x$.

2

(ii) Hence or otherwise, find a primitive of $\frac{x + 1}{x^2 + x \log_e x}$.

3

(c) AKB , CKD are two chords of a circle (meeting at an internal point K).

3

Given the following lengths

$AB = 10$ cm, $CD = 6$ cm, $AK = 1$ cm,

calculate the ratio $AC : BD$.

Question 6 (12 points)

- (a) (i) How many three-figure numbers can be formed from the nine digits 1, 2, 3, ..., 9, there being no repetitions? 1
- (ii) From a pack of nine cards numbered 1, ..., 9, three cards are drawn at random and laid on a table from left to right.
- (α) What is the probability that the number formed by the three digits drawn should exceed 500? 1
- (β) What is the probability that the digits should be drawn in ascending order, not necessarily consecutive? 3
- (b) (i) Prove that the graph of $y = \ln x$ is concave down for all $x > 0$. 2
- (ii) Sketch the graph of $y = \ln x$. 1
- (iii) Suppose $1 < a < b$ and consider the points $A(a, \ln a)$ and $B(b, \ln b)$ on the graph of $y = \ln x$. 2
- Find the coördinates of the point P that divides the line segment AB in the ratio 2 : 1.
- (iv) By using (ii) and (iii), deduce that $\frac{1}{3} \ln a + \frac{2}{3} \ln b < \ln \left(\frac{1}{3}a + \frac{2}{3}b \right)$. 2

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

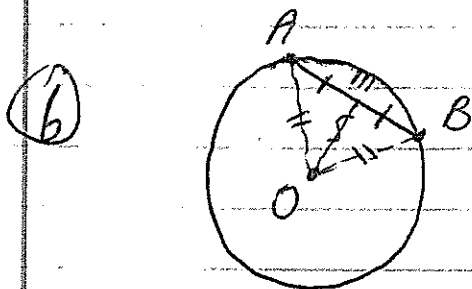
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

(1)(a)(i) $4 \times 3 \times 2 \times 1 = 4! = 24$ ①

(ii) $\underline{3} \times \underline{2} \times \underline{1} \times \underline{3} = 18$ ①

12.



Join AO and BO
In $\triangle AOM$ and $\triangle BOM$

AO = BO radii

AM = MB data

OM common side

$\therefore \triangle AOM \equiv \triangle BOM$ (SSS)

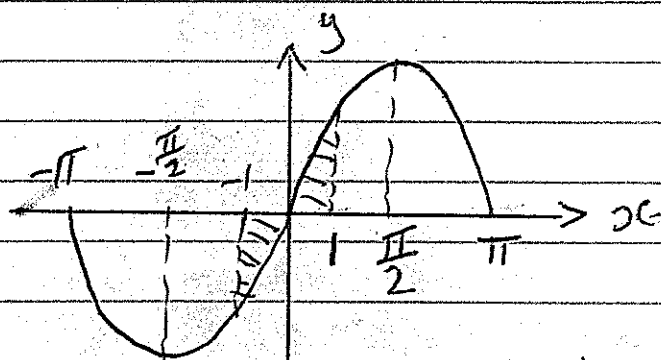
Hence $\hat{AMO} + \hat{BMO} = 180^\circ$

But $\hat{AMO} = \hat{BMO}$ corresponding angles in congruent triangles

So $\hat{AMO} = \hat{BMO} = 90^\circ$

Hence line from centre of circle to the midpt of a chord is $\perp$ to the chord. ③

(c)



area under curve (shaded) cancels out area above curve (shaded)

$\sin x$ is an odd function

②

(2)

$$(c) (ii) \lim_{x \rightarrow 0} \frac{\tan 4x}{7x}$$

$$\frac{4}{7} \lim_{x \rightarrow 0} \frac{\tan 4x}{4x} = \frac{4}{7} \times 1 = \frac{4}{7} \quad (1)$$

$$(d) (i) \frac{d}{dx} (x \ln x) = x \times \frac{1}{x} + \ln x \times 1$$

$$= 1 + \ln x \quad (1)$$

$$\text{so } \frac{d}{dx} (x \ln x) - 1 = \ln x$$

$$(ii) \int \left(\frac{d}{dx} (x \ln x) - 1 \right) dx = \int \ln x \, dx$$

$$\therefore \int \ln x \, dx = x \ln x - x + C \quad (2)$$

$$(e) 2 \times (10+9+8+7+6+5+4+3+2) = 110 \quad (1)$$

or ${}^{11}P_2$

$$(2) (a) (i) \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \tan^2 x \, dx \quad \text{using } \tan^2 x = \sec^2 x - 1$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (\sec^2 x - 1) \, dx$$

$$= \left(\tan x - x \right) \Big|_{\frac{\pi}{6}}^{\frac{\pi}{3}} \quad 12.$$

$$= \left(\tan \frac{\pi}{3} - \frac{\pi}{3} \right) - \left(\tan \frac{\pi}{6} - \frac{\pi}{6} \right)$$

$$= \sqrt{3} - \frac{\pi}{3} - \frac{1}{\sqrt{3}} + \frac{\pi}{6} = \frac{2}{\sqrt{3}} - \frac{\pi}{6} = \left(\frac{2\sqrt{3}}{3} - \frac{\pi}{6} \right) \quad (2)$$

2 (a) (ii) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin^2 x \, dx$

$$= \left(\frac{1}{2}x - \frac{1}{4}\sin 2x \right)_{\frac{\pi}{6}}^{\frac{\pi}{3}}$$

$$= \left(\frac{\pi}{6} - \frac{1}{4}\sin \frac{2\pi}{3} \right) - \left(\frac{\pi}{12} - \frac{1}{4}\sin \frac{\pi}{3} \right)$$

$$= \frac{\pi}{6} - \frac{\pi}{12} - \frac{1}{4} \times \frac{\sqrt{3}}{2} + \frac{1}{4} \times \frac{\sqrt{3}}{2}$$

$$= \frac{\pi}{12} \quad (2)$$

(b) (i) m. Induction $\sum_{r=1}^n \frac{1}{r(r+1)} = \frac{n}{n+1}$

LHS means: $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$
 $r=1$ $r=2$ $r=3$ sum

Step 1 let $r=1$, LHS = $\frac{1}{1 \times 2} = \frac{1}{2}$

RHS, $n=1$ term $\frac{1}{1+1} = \frac{1}{2}$
 True for $n=1$.

Step 2 let there be a value of $n=k$, ($k \leq n$) such that $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$ is true

and we must prove that when $n=k+1$

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = \frac{k+1}{k+2}$$

LHS. $\frac{k}{(k+1)} + \frac{1}{(k+1)(k+2)} = \frac{k(k+2) + 1}{(k+1)(k+2)} = \frac{(k+1)^2}{(k+1)(k+2)}$

(4)

$$\text{LHS } \frac{(k+1)}{(k+2)}$$

$$= \text{RHS}$$

$\therefore n = k+1$ is true

(4)

Step 3 By the principle of math. induction if it is true for $n=k$, it is true for $n=k+1$ etc.

(b) (ii) as $n \rightarrow \infty$, $\lim_{n \rightarrow \infty} \left(\sum_{r=1}^n \frac{1}{r(r+1)} \right) = 1$

(1)

2 (c) $y = \sin^3 x - 0.25$
 $y' = 3\sin^2 x \cos x$

Using
 $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$

$$y = \sin^3 0.7 - 0.25 = 0.01736$$

$$y' = 3\sin^2 0.7 \times \cos 0.7 = \cancel{\text{WATER}} 0.95227$$

$$x_1 = 0.7 - \frac{0.01736}{\cancel{\text{WATER}} 0.95227}$$

$$\approx \cancel{\text{WATER}} (3DP) 0.682$$

(3)

Extension 1 Solutions

SECTION B

Q3 (a) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \cot x \, dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\cos x}{\sin x} \, dx$

$$= \left[\ln \sin x \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$$
$$= \ln \sin \frac{\pi}{3} - \ln \sin \frac{\pi}{6}$$
$$= \ln \frac{\sqrt{3}}{2} - \ln \frac{1}{2}$$
$$= \ln \left(\frac{\sqrt{3}}{2} \div \frac{1}{2} \right) \text{ [using log laws]}$$
$$= \ln \sqrt{3}$$
$$= 0.549 \text{ correct to 3 decimal places}$$

(b)(i) Having at least one of the two best players means having one of the two or both of them

$$\therefore \text{No. teams} = {}^2C_1 \times {}^8C_4 + {}^2C_2 \times {}^8C_3$$
$$= 140 + 56$$
$$= 196.$$

(ii) Having no more than one of the three youngest players means having none of the youngest players or having just one of them

$$\therefore \text{No. teams} = {}^7C_5 + {}^3C_1 \times {}^7C_4$$
$$= 21 + 105$$
$$= 126.$$

13/ (c) (i) Let $f(x) = e^x - 3x - 2$ and consider $f(x) = 0$.
 Now $f(2) = e^2 - 6 - 2 = 7.4 - 6 - 2 = -0.6$
 and $f(2.5) = e^{2.5} - 7.5 - 2 = 12.18 - 9.5 = 2.68$

Now $f(x)$ is continuous for $2 \leq x \leq 2.5$ and changes sign in that interval.

$\therefore f(x)$ has a zero between 2 and 2.5.

(ii) Consider $f\left(\frac{2+2.5}{2}\right) = f(2.25) = 9.488 - 8.75 = 0.734$

$\therefore$ There is a root between $x=2$ and $x=2.25$

Consider $f\left(\frac{2+2.25}{2}\right) = f(2.125) = e^{2.125} - 8.375 = -0.001$

$\therefore$ There is a root between $x=2.125$ and $x=2.25$

Consider $f\left(\frac{2.125+2.25}{2}\right) = f(2.188) \doteq 0.349$

$\therefore$ There is a root between $x=2.188$ and $x=2.125$

Consider $f\left(\frac{2.188+2.125}{2}\right) = f(2.157) \doteq 0.173$

$\therefore$ There is a root between $x=2.157$ and $x=2.125$

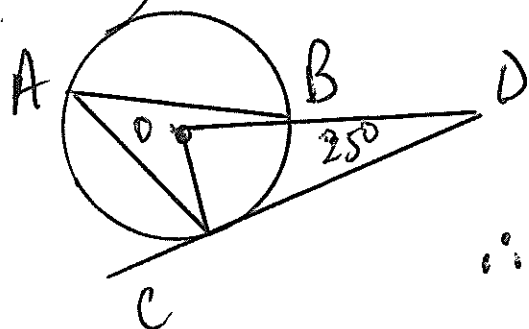
Consider $f\left(\frac{2.157+2.125}{2}\right) = f(2.141) \doteq 0.081$

$\therefore$ There is a root between $x=2.125$ and $x=2.141$

And 2.125 and 2.141 are both 2.1 correct to one decimal place.

$\therefore$ No further application of halving the interval is required to find the required root
 i.e. The root is 2.1.

Q4(a)



Examine the figure as labelled.

$$\angle OCD = 90^\circ \left[\begin{array}{l} \text{tangent } \perp \text{ to radius} \\ \text{drawn to point of contact} \end{array} \right]$$

$$\therefore \angle COD = 180 - (90 + 25)^\circ \left[\begin{array}{l} \text{sum of } \Delta \\ \text{is } 180^\circ \end{array} \right]$$

$$= 65^\circ$$

$$\therefore \angle CAB = \frac{65^\circ}{2} \left[\begin{array}{l} \text{angle at centre } 2 \times \text{angle} \\ \text{at circumference on same} \\ \text{arc} \end{array} \right]$$

$$= 32.5^\circ$$

(b) Consider $S(n) = 7^{n+1} + 3^n$

Step 1: Let $n=1$, then $S(1) = 7^{1+1} + 3^1 = 49 + 3 = 52$

$$\text{And } 52 = 4 \times 13$$

$\therefore$ The statement is true for $n=1$. — (A)

Step 2: Assume that the statement is true for $n=k$

i.e. Assume that $7^{k+1} + 3^k = 4A$ (A an integer)

Now consider the statement for $n=k+1$.

$$\text{Then } S(k+1) = 7^{k+2} + 3^{k+1}$$

$$= 7 \cdot 7^{k+1} + 3^{k+1}$$

$$= 4 \cdot 7^{k+1} + 3 \cdot 7^{k+1} + 3 \cdot 3^k$$

$$= 4 \cdot 7^{k+1} + 3(7^{k+1} + 3 \cdot 3^k)$$

$$= 4 + 3 \cdot 4A \text{ using the assumption}$$

$$= 4(1 + 3A) \text{ and } 1 + 3A \text{ is integral}$$

$\therefore$ The statement is true for $n=k+1$ if it is true for $n=k$ as assumed.

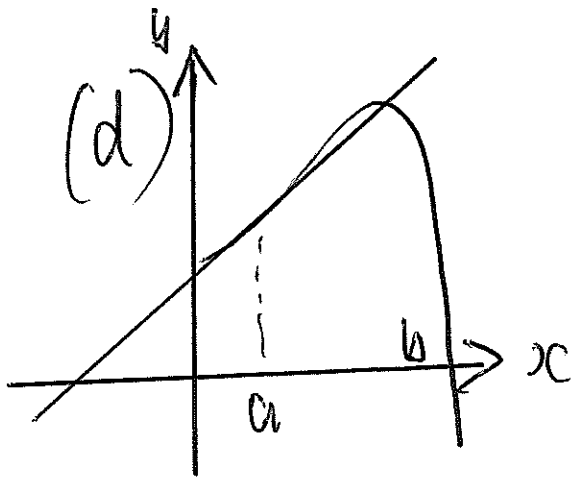
Step 3: But the statement is true for $n=1$ (from A)

$\therefore$ True for $n=1+1=2 \rightarrow$ true for $2+1=3$
and so on for all integral n .

04 (c) No. of teams possible without restriction $= {}^6C_3$
 $= 165$

No. of teams with mixed sex $=$ No. teams $2M \times 1W$ $+$ No Teams $1M \times 2W$
 $= {}^6C_2 \times {}^5C_1 + {}^6C_1 \times {}^5C_2$
 $= 75 + 60$
 $= 135.$

$\therefore P(E) = \frac{135}{165}$
 $= \frac{9}{11}.$



Newton's method would not give a better approximation because there is a turning point between a and b .

Thus the tangent to the curve at a will cut the x -axis further away from b than a is.

Section C

5)a) i) $y = \log_e(\cos x)$

$$y' = \frac{-\sin x}{\cos x}$$

$$y' = -\tan x$$

(ii) $y = (x+1)e^{-x}$

$$\begin{array}{l} u = x+1 \\ u' = 1 \end{array} \quad \times \quad \begin{array}{l} v = e^{-x} \\ v' = -e^{-x} \end{array}$$

$$y' = -(x+1)e^{-x} + e^{-x}$$

$$y' = -xe^{-x} - e^{-x} + e^{-x}$$

$$y' = -xe^{-x}$$

(b) i) $\frac{d(x + \log_e x)}{dx} = 1 + \frac{1}{x}$

$$= \frac{x+1}{x}$$

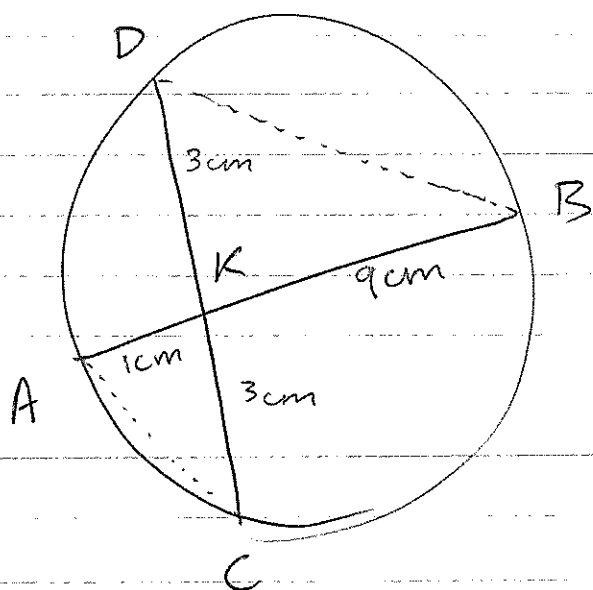
(ii) $\int \frac{x+1}{x^2 + x \log_e x} dx = \int \frac{x+1}{x(x + \log_e x)} dx$

$$= \int \frac{\left(\frac{x+1}{x}\right)}{x + \log_e x} dx$$

in form $\int \frac{f'(x)}{f(x)} dx$

$$= \log_e(x + \log_e x) + C$$

(c)



$$CD = 6 \text{ cm.}$$

$AK \cdot KB = CK \cdot KD$
(product of intercepts,)
intersecting chords

$$\text{let } CK = x$$

$$1 \times 9 = x(6 - x)$$

$$6x - x^2 = 9$$

$$x^2 - 6x + 9 = 0$$

$$(x - 3)^2 = 0$$

$$x = 3.$$

In Δ 's AKC & DKB

$\angle AKC = \angle DKB$ (vertically opposite angles)

$\angle CAK = \angle BDK$ (angles in same segment)

$\therefore \Delta AKC \parallel \Delta DKB$ (equiangular)

$$\frac{AC}{BD} = \frac{AK}{DK} \quad (\text{corresponding sides of similar triangles in same ratio})$$

$$\frac{AC}{BD} = \frac{1}{3}$$

$$\therefore AC : BD = 1 : 3$$

$$6) a) i) \quad 9 \times 8 \times 7 = 504 \quad \text{or } {}^9P_3$$

$$(ii) \times) \quad 5 \times 8 \times 7 = 280$$

$$\text{Probability} = \frac{280}{504}$$

$$= \frac{5}{9}$$

(B) Given a 3 digit number (of different digits not including zero), only one arrangement will be in ascending order.

$$\text{ie } \frac{1}{3!} = \frac{1}{6}$$

$$\text{or } \frac{{}^3C_3}{{}^3P_3}$$

(b) i) $y = \ln x$, $x > 0$

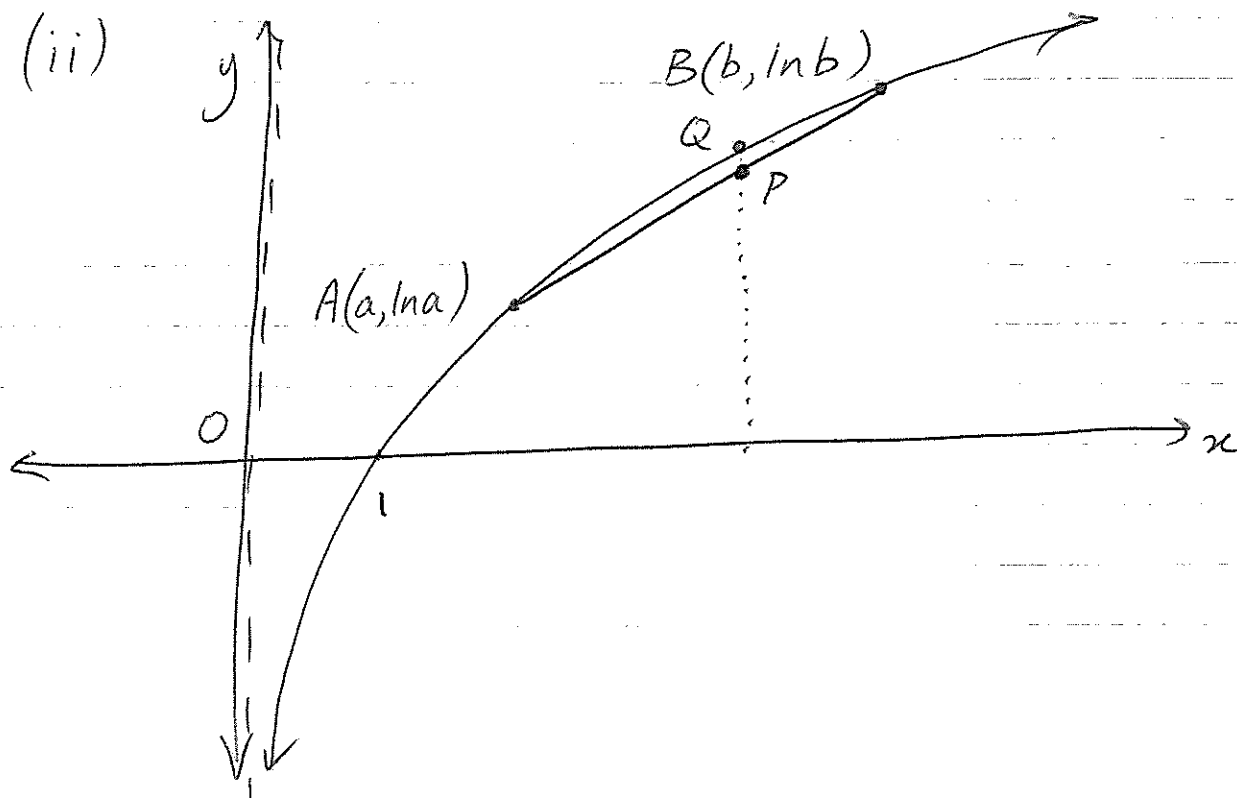
$$y' = \frac{1}{x} = x^{-1}$$

$$y'' = -x^{-2}$$

$$= -\frac{1}{x^2}$$

$$y'' < 0 \quad \text{for all } x > 0$$

$\therefore y = \ln x$ is concave down for all $x > 0$.



(iii) $A(a, \ln a)$ $\xrightarrow{2:1}$ $B(b, \ln b)$

$$P = \left(\frac{a + 2b}{2 + 1}, \frac{\ln a + 2 \ln b}{2 + 1} \right)$$

$$P = \left(\frac{1}{3}a + \frac{2}{3}b, \frac{1}{3}\ln a + \frac{2}{3}\ln b \right)$$

(iv) Consider the point Q which lies on $y = \ln x$ with the same x -coordinate as P .

Clearly from the graph in (ii)

$$y_P < y_Q.$$

$$\frac{1}{3}\ln a + \frac{2}{3}\ln b < \ln\left(\frac{1}{3}a + \frac{2}{3}b\right)$$